

Practical 4

Aim:

To process the NCDC historical weather dataset using Hadoop MapReduce and determine the maximum temperature recorded for each year, analyze yearly temperature trends, and identify the hottest year in the dataset.

Problem definition:

A meteorological research organization maintains a large collection of historical weather records collected from weather stations worldwide. Processing such large datasets using traditional sequential methods can be time-consuming and inefficient.

The objective of this practical is to use the Hadoop MapReduce framework to process the NCDC weather dataset. The Mapper extracts the **year** and **temperature** from each weather record and generates intermediate key-value pairs. The Reducer processes all temperature values belonging to the same year and determines the maximum temperature recorded during that year.

The final output is stored in HDFS and used to analyze long-term temperature trends and identify the hottest year in the dataset.

Objectives:

1. To upload the NCDC weather dataset into HDFS.
2. To examine and understand the structure of NCDC weather records.
3. To develop a Mapper program for extracting the year and temperature values.
4. To generate intermediate key-value pairs in the form (Year, Temperature).
5. To develop a Reducer program for finding the maximum temperature for each year.
6. To execute the MapReduce job on Hadoop.
7. To store and verify the generated output in HDFS.
8. To analyze yearly maximum temperature trends.
9. To identify the hottest year in the dataset.
10. To understand the advantages of MapReduce for large-scale climate data analysis.

Observation / Output:

Task 1: Upload the NCDC weather dataset to HDFS and examine the structure of the weather records.

```
hdfs dfs -put ncdc_weather.txt /user/tushar/ncdc/input/  
hdfs dfs -ls /user/tushar/ncdc/input  
hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt
```

```

tushar@Tushar-Laptop: ~
$ nano ncdc_weather.txt
$ cat ncdc_weather.txt
US000094728 20200101 32
US000094728 20200102 28
US000094728 20200715 35
US000094728 20200720 36
US000094728 20210110 30
US000094728 20210615 36
US000094728 20210620 39
US000094728 20220105 29
US000094728 20220718 41
US000094728 20220810 31
US000094728 20230815 43
US000094728 20240112 30
US000094728 20240620 42

$ wc -l ncdc_weather.txt
13 ncdc_weather.txt

$ hdfs dfs -mkdir -p /user/tushar/ncdc/input

$ hdfs dfs -ls /user/tushar/ncdc
Found 1 items
drwxr-xr-x  - tushar supergroup          0 2026-09-14 17:26 /user/tushar/ncdc/input

$ hdfs dfs -put ncdc_weather.txt /user/tushar/ncdc/input/

$ hdfs dfs -ls /user/tushar/ncdc/input
Found 1 items
-rw-r--r--  1 tushar supergroup        312 2026-09-14 17:26 /user/tushar/ncdc/input/ncdc_weather.txt

$ hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt
US000094728 20200101 32
US000094728 20200102 28
US000094728 20200715 35
US000094728 20200720 36
US000094728 20210110 30
US000094728 20210615 36
US000094728 20210620 39
US000094728 20220105 29
US000094728 20220718 41
US000094728 20220810 31
US000094728 20230815 43
US000094728 20240112 30
US000094728 20240620 42
  
```

Task 2: Develop and implement the Mapper program to extract the year and temperature values from each record and generate intermediate key-value pairs.

```

cd ~/ncdc_project
nano mapper.py
chmod +x mapper.py
hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt | ./mapper.py
  
```

```

tushar@Tushar-Laptop: ~
$ hdfs dfs -ls /user/tushar/ncdc
Found 1 items
drwxr-xr-x  - tushar supergroup          0 2026-09-14 17:26 /user/tushar/ncdc/input

$ hdfs dfs -put ncdc_weather.txt /user/tushar/ncdc/input/

$ hdfs dfs -ls /user/tushar/ncdc/input
Found 1 items
-rw-r--r--  1 tushar supergroup        312 2026-09-14 17:26 /user/tushar/ncdc/input/ncdc_weather.txt

$ hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt
US000094728 20200101 32
US000094728 20200102 28
US000094728 20200715 35
US000094728 20200720 36
US000094728 20210110 30
US000094728 20210615 36
US000094728 20210620 39
US000094728 20220105 29
US000094728 20220718 41
US000094728 20220810 31
US000094728 20230815 43
US000094728 20240112 30
US000094728 20240620 42

$ cd ~/ncdc_project
$ nano mapper.py
$ chmod +x mapper.py

$ hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt | ./mapper.py
2020 32
2020 28
2020 35
2020 36
2021 30
2021 36
2021 39
2022 29
2022 41
2023 31
2023 43
2024 30
2024 42
  
```

Task 3: Develop and implement the Reducer program to determine the maximum temperature recorded for each year.

```

cd ~/ncdc_project
nano reducer.py
hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt | ./mapper.py | sort | ./reducer.py
  
```

```

tushar@Tushar-laptop: ~
Session Actions Edit View Help
USW00094728 20210110 30
USW00094728 20210615 36
USW00094728 20210609 39
USW00094728 20210805 29
USW00094728 20220718 41
USW00094728 20220810 51
USW00094728 20230815 43
USW00094728 20240112 30
USW00094728 20240620 42

tushar@Tushar-laptop: ~
^C

tushar@Tushar-laptop: ~
cd ~/ncdc_project
bash: cd: /home/tushar/ncdc_project: No such file or directory

tushar@Tushar-laptop: ~
nano mapper.py

tushar@Tushar-laptop: ~
chmod +x mapper.py

tushar@Tushar-laptop: ~
hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt | ./mapper.py
2020 32
2020 26
2020 25
2020 38
2021 30
2021 36
2021 39
2022 29
2022 41
2023 31
2023 43
2024 30
2024 42

tushar@Tushar-laptop: ~
cd ~/ncdc_project
bash: cd: /home/tushar/ncdc_project: No such file or directory

tushar@Tushar-laptop: ~
nano reducer.py

tushar@Tushar-laptop: ~
chmod +x reducer.py

tushar@Tushar-laptop: ~
hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt | ./mapper.py | sort | ./reducer.py
2020 38
2021 39
2022 41
2023 43
2024 42

tushar@Tushar-laptop: ~

```

Task 4: Execute the MapReduce job on Hadoop and store the results in HDFS. Verify the correctness of the output generated.

```
hdfs dfs -cat /user/tushar/ncdc/output/part-00000
```

```
hdfs dfs -ls /user/tushar/ncdc/output
```

```
hdfs dfs -get /user/tushar/ncdc/output/part-00000 ~/yearly_max_temperature.txt
```

```

tushar@Tushar-laptop: ~
Session Actions Edit View Help

tushar@Tushar-laptop: ~
cd ~/ncdc_project
bash: cd: /home/tushar/ncdc_project: No such file or directory

tushar@Tushar-laptop: ~
hdfs dfs -ls /user/tushar/ncdc/output
rm: /user/tushar/ncdc/output: No such file or directory

tushar@Tushar-laptop: ~
ls $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar
/home/tushar/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.4.3.jar

tushar@Tushar-laptop: ~
hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
> -input /user/tushar/ncdc/input/ncdc_weather.txt \
> -output /user/tushar/ncdc/output \
> -mapper ~/ncdc_project/mapper.py \
> -reducer ~/ncdc_project/reducer.py
packageJobJar: [/tmp/hadoop-unjar96875392284635943/] [] /tmp/streamjob1602955788940233859.jar tmpDir=null
2026-09-14 17:40:36,624 INFO client.DefaultHadoopFollowerProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-14 17:40:36,696 INFO client.DefaultHadoopFollowerProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-14 17:40:37,555 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/tushar/.staging/job_1789386439716_0001
2026-09-14 17:40:37,963 INFO mapred.FileInputFormat: Total input files to process : 1
2026-09-14 17:40:38,077 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-14 17:40:38,374 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789386439716_0001
2026-09-14 17:40:38,375 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-14 17:40:38,658 INFO conf.Configuration: resource-types.xml not found
2026-09-14 17:40:38,631 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-14 17:40:39,096 INFO impl.YarnClientImpl: Submitted application application_1789386439716_0001
2026-09-14 17:40:39,167 INFO mapreduce.Job: The url to track the job: http://tushar-laptop.localdomain:8888/proxy/application_1789386439716_0001/
2026-09-14 17:40:39,169 INFO mapreduce.Job: Running job: job_1789386439716_0001
2026-09-14 17:40:41,238 INFO mapreduce.Job: Job job_1789386439716_0001 running in uber mode : false
2026-09-14 17:40:47,340 INFO mapreduce.Job: map 0% reduce 0%
2026-09-14 17:40:52,492 INFO mapreduce.Job: Task Id : attempt_1789386439716_0001_m_0000000_0, Status : FAILED
Error: java.lang.RuntimeException: Error in configuring object
    at org.apache.hadoop.util.ReflectionUtils.setJobConf(ReflectionUtils.java:113)
    at org.apache.hadoop.util.ReflectionUtils.setConf(ReflectionUtils.java:79)
    at org.apache.hadoop.util.ReflectionUtils.newInstance(ReflectionUtils.java:159)
    at org.apache.hadoop.mapred.MapTask.run(WrapperMapTask.java:463)
    at org.apache.hadoop.mapred.MapTask.run(WrapperMapTask.java:463)
    at org.apache.hadoop.mapred.YarnChild$2.run(YarnChild.java:178)
    at java.base/java.security.auth.Subject.doAs(Subject.java:379)
    at java.base/java.security.AccessController.doPrivileged(AccessController.java:714)
    at java.base/java.security.auth.Subject.doAs(Subject.java:525)
    at java.base/java.security.auth.Subject.doAs(Subject.java:381)
    at org.apache.hadoop.security.authentication.util.SubjectUtil.callAs(SubjectUtil.java:161)
    at org.apache.hadoop.security.authentication.util.SubjectUtil.doAs(SubjectUtil.java:233)
    at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1951)
    at org.apache.hadoop.mapred.YarnChild.main(YarnChild.java:172)
Caused by: java.lang.reflect.InvocationTargetException
    at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:118)
    at java.base/java.lang.reflect.Method.invoke(Method.java:580)
    at org.apache.hadoop.util.ReflectionUtils.setJobConf(ReflectionUtils.java:110)
    ... 16 more
Caused by: java.lang.RuntimeException: Error in configuring object
    at org.apache.hadoop.util.ReflectionUtils.setJobConf(ReflectionUtils.java:113)
    at org.apache.hadoop.util.ReflectionUtils.setConf(ReflectionUtils.java:79)
    at org.apache.hadoop.util.ReflectionUtils.newInstance(ReflectionUtils.java:159)

```

```

tushar@Tushar-laptop: ~
Session Actions Edit View Help

hdfs dfs -rm -r /user/tushar/ncdc/output
rm: /user/tushar/ncdc/output: No such file or directory

tushar@Tushar-laptop: ~
$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-3.4.3.jar \
> -files mapper.py, reducer.py \
> -input /user/tushar/ncdc/output/ncdc_weather.txt \
> -output /user/tushar/ncdc/output \
> -mapper "python3 mapper.py" \
> -reducer "python3 reducer.py" \
packageJobJar: [/tmp/hadoop-unjar4486367301820435992/] [] /tmp/streamjob9677462674120931660.jar tmpDir=null
2026-09-14 17:46:30.420 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-14 17:46:30.529 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-14 17:46:30.813 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/tushar/.staging/job_1789386439716_0002
2026-09-14 17:46:31.154 INFO mapreduce.JobInputFormat: Total Input Files to process : 1
2026-09-14 17:46:31.210 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-14 17:46:31.353 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789386439716_0002
2026-09-14 17:46:31.353 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-14 17:46:31.521 INFO conf.Configuration: resource-types.xml not found
2026-09-14 17:46:31.521 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-14 17:46:31.521 INFO impl.YarnClientImpl: Submitted application application_1789386439716_0002
2026-09-14 17:46:31.628 INFO mapreduce.Job: The url to track the job: http://Tushar-laptop.localdomain:8088/proxy/application_1789386439716_0002/
2026-09-14 17:46:31.629 INFO mapreduce.Job: Running Job: job_1789386439716_0002
2026-09-14 17:46:37.723 INFO mapreduce.Job: Job Job_1789386439716_0002 running in uber mode : false
2026-09-14 17:46:37.724 INFO mapreduce.Job: map 0% reduce 0%
2026-09-14 17:46:44.811 INFO mapreduce.Job: map 100% reduce 0%
2026-09-14 17:46:46.852 INFO mapreduce.Job: map 100% reduce 100%
2026-09-14 17:46:46.861 INFO mapreduce.Job: Job Job_1789386439716_0002 completed successfully
2026-09-14 17:46:46.941 INFO mapreduce.Job: Counters: 54

File System Counters
  FILE: Number of bytes read=136
  FILE: Number of bytes written=942991
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=694
  HDFS: Number of bytes written=40
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0

Job Counters
  Launched map tasks=2
  Launched reduce tasks=1
  Data-Local map tasks=2
  Total time spent by all maps in occupied slots (ms)=5087
  Total time spent by all reduces in occupied slots (ms)=2412
  Total time spent by all map tasks (ms)=5087
  Total time spent by all reduce tasks (ms)=2412
  Total vcore-milliseconds taken by all map tasks=5087
  Total vcore-milliseconds taken by all reduce tasks=2412
  Total megabyte-milliseconds taken by all map tasks=5209088
  Total megabyte-milliseconds taken by all reduce tasks=2469888

Map-Reduce Framework
  Map input records=13
  Map output records=13
  Map output bytes=184
  Map output materialized bytes=142
  Input split bytes=220
  Combine input records=0
  Combine output records=0
  Reduce input groups=5
  Reduce shuffle bytes=142
  Reduce input records=12
  Reduce output records=5
  Spilled Records=26
  Shuffled Maps=2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=66
  CPU time spent (ms)=1870
  Physical memory (bytes) snapshot=888251392
  Virtual memory (bytes) snapshot=8386576384
  Total committed heap usage (bytes)=440481920
  Peak Map Physical memory (bytes)=336090376
  Peak Map Virtual memory (bytes)=2795229184
  Peak Reduce Physical memory (bytes)=197181440
  Peak Reduce Virtual memory (bytes)=2797078392

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=608
  Bytes Written=40

2026-09-14 17:46:46.941 INFO streaming.StreamJob: Output directory: /user/tushar/ncdc/output

tushar@Tushar-laptop: ~
$ hdfs dfs -ls /user/tushar/ncdc/output
Found 2 Items
-rw-r--r-- 1 tushar supergroup 0 2026-09-14 17:46 /user/tushar/ncdc/output/_SUCCESS
-rw-r--r-- 1 tushar supergroup 40 2026-09-14 17:46 /user/tushar/ncdc/output/part-000000

tushar@Tushar-laptop: ~
$ hdfs dfs -cat /user/tushar/ncdc/output/part-000000
2020 38
2021 39
2022 41
2023 43
2024 42

```

```

tushar@Tushar-laptop: ~
Session Actions Edit View Help

Total time spent by all reduces in occupied slots (ms)=2412
Total time spent by all map tasks (ms)=5087
Total time spent by all reduce tasks (ms)=2412
Total vcore-milliseconds taken by all map tasks=5087
Total vcore-milliseconds taken by all reduce tasks=2412
Total megabyte-milliseconds taken by all map tasks=5209088
Total megabyte-milliseconds taken by all reduce tasks=2469888

Map-Reduce Framework
  Map input records=13
  Map output records=13
  Map output bytes=184
  Map output materialized bytes=142
  Input split bytes=220
  Combine input records=0
  Combine output records=0
  Reduce input groups=5
  Reduce shuffle bytes=142
  Reduce input records=12
  Reduce output records=5
  Spilled Records=26
  Shuffled Maps=2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=66
  CPU time spent (ms)=1870
  Physical memory (bytes) snapshot=888251392
  Virtual memory (bytes) snapshot=8386576384
  Total committed heap usage (bytes)=440481920
  Peak Map Physical memory (bytes)=336090376
  Peak Map Virtual memory (bytes)=2795229184
  Peak Reduce Physical memory (bytes)=197181440
  Peak Reduce Virtual memory (bytes)=2797078392

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=608
  Bytes Written=40

2026-09-14 17:46:46.941 INFO streaming.StreamJob: Output directory: /user/tushar/ncdc/output

tushar@Tushar-laptop: ~
$ hdfs dfs -ls /user/tushar/ncdc/output
Found 2 Items
-rw-r--r-- 1 tushar supergroup 0 2026-09-14 17:46 /user/tushar/ncdc/output/_SUCCESS
-rw-r--r-- 1 tushar supergroup 40 2026-09-14 17:46 /user/tushar/ncdc/output/part-000000

tushar@Tushar-laptop: ~
$ hdfs dfs -cat /user/tushar/ncdc/output/part-000000
2020 38
2021 39
2022 41
2023 43
2024 42

```

Task 5: Analyze the yearly maximum temperature trends and identify the hottest year in the dataset. Discuss the advantages of using MapReduce for large-scale climate data analysis.

```

tushar@Tushar-laptop: ~
$ hdfs dfs -cat /user/tushar/ncdc/output/part-000000
2020 38
2021 39
2022 41
2023 43
2024 42

tushar@Tushar-laptop: ~
$

```

hdfs dfs -cat /user/tushar/ncdc/output/part-000000

Advantages of MapReduce for large-scale climate data analysis:

- **Parallel Processing:** Processes large amounts of climate data simultaneously across multiple machines.
- **Scalability:** Can handle very large datasets by adding more nodes.

- **Fault Tolerance:** Automatically handles machine failures and reprocesses lost data.
- **Distributed Storage:** HDFS stores huge climate datasets across multiple machines.
- **Faster Processing:** Map and Reduce tasks work in parallel, making large-scale analysis faster.
- **Data Locality:** Processes data close to where it is stored, reducing network transfer.

Key Questions /Analysis:

Q1. How is weather data parsed and processed using MapReduce?

Weather data is first stored in HDFS. The Mapper reads each weather record, extracts the year and temperature, and generates key-value pairs in the form (year, temperature). The Hadoop framework then groups all temperatures belonging to the same year. The Reducer processes these grouped values and finds the maximum temperature for each year. Finally, the yearly maximum temperatures are stored as the MapReduce output in HDFS.

Example:

Input: USW00094728 20230815 43

Mapper: 2023 43

Reducer: 2023 43

Thus, MapReduce efficiently parses and processes large weather datasets by distributing the computation across multiple nodes.

Q2. What key-value pairs are generated by the Mapper?

The Mapper extracts the year from the date and the temperature from each weather record. It generates the year as the key and temperature as the value.

Example:

2020 32

2020 28

2020 35

2020 38

2021 30

2021 36

2021 39

2022 29

Thus, the Mapper generates (Year, Temperature) key-value pairs, which are then passed to the Reducer for finding the maximum temperature for each year.

Q3. How does the Reducer identify the maximum temperature for a given year?

The Reducer receives all temperature values associated with the same year. It compares the temperatures one by one and keeps track of the highest value. After processing all values for that year, it outputs the year along with its maximum temperature.

Example:

2023 → 31, 43
Maximum = 43°C

Therefore, the Reducer outputs:

2023 43

Thus, the Reducer identifies the maximum temperature by comparing all temperature values belonging to each year.

Q4. What challenges arise when processing large climate datasets?

Processing large climate datasets can involve the following challenges:

- Huge Data Volume: Climate data can contain millions or billions of records.
- High Processing Time: Processing such large datasets on a single machine can be slow.
- Storage Requirements: Large datasets require significant storage capacity.
- Data Quality: Missing, incorrect, or inconsistent weather records can affect the analysis.
- Scalability: Traditional systems may struggle as the dataset continues to grow.
- Fault Handling: Hardware or system failures can interrupt processing of large datasets.

MapReduce helps overcome these challenges by providing distributed storage, parallel processing, scalability, and fault tolerance.

Q5. How does Hadoop improve scalability and performance for weather data analytics?

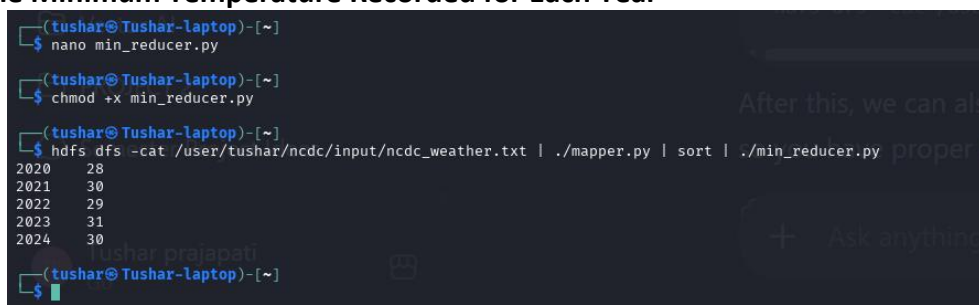
Hadoop improves scalability and performance by distributing large weather datasets and computations across multiple machines in a cluster.

- Distributed Storage: HDFS stores large weather datasets across multiple nodes.
- Parallel Processing: MapReduce processes different portions of data simultaneously.
- Scalability: More nodes can be added to handle increasing amounts of climate data.
- Fault Tolerance: Hadoop can continue processing even if a node fails.
- Data Locality: Hadoop processes data close to where it is stored, reducing data transfer.
- Efficient Processing: Large datasets can be analyzed faster compared to processing everything on a single machine.

Hadoop provides a scalable, reliable, and efficient platform for analyzing large-scale weather and climate datasets.

Supplementary Problems:

1. Find the Minimum Temperature Recorded for Each Year



```
(tushar@Tushar-laptop)~$ nano min_reducer.py
(tushar@Tushar-laptop)~$ chmod +x min_reducer.py
(tushar@Tushar-laptop)~$ hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt | ./mapper.py | sort | ./min_reducer.py
2020 28
2021 30
2022 29
2023 31
2024 30
(tushar@Tushar-laptop)~$
```

The terminal output shows the results of the MapReduce job, displaying the minimum temperature for each year from 2020 to 2024. The results are: 2020 (28), 2021 (30), 2022 (29), 2023 (31), and 2024 (30).

2. Calculate Average Yearly Temperatures

```
(tushar@Tushar-laptop)-[~]
$ nano avg_reducer.py

(tushar@Tushar-laptop)-[~]
$ chmod +x avg_reducer.py

(tushar@Tushar-laptop)-[~]
$ hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt | ./mapper.py | sort | ./avg_reducer.py
2020    33.25
2021    35.00
2022    35.00
2023    37.00
2024    36.00

(tushar@Tushar-laptop)-[~]
$
```

3. Identify the Year with the Highest Recorded Temperature Overall

```
(tushar@Tushar-laptop)-[~]
$ chmod +x overall_max_reducer.py

(tushar@Tushar-laptop)-[~]
$ hdfs dfs -cat /user/tushar/ncdc/input/ncdc_weather.txt | ./mapper.py | sort | ./overall_max_reducer.py
2023    43

(tushar@Tushar-laptop)-[~]
$
```

Tools / Technology used:

- **Hadoop:** Hadoop 3.x
- **Distributed Storage:** HDFS
- **Processing Framework:** MapReduce
- **Programming Language:** Java
- **Java Version:** JDK 8 / JDK 11
- **Operating System:** Linux / Ubuntu
- **Development Environment:** Eclipse / IntelliJ IDEA / Terminal
- **Resource Management:** YARN
- **Dataset:** NCDC Historical Weather Dataset

Conclusion

The NCDC weather data was successfully processed using Hadoop MapReduce with Python Hadoop Streaming. The Mapper extracted the year and temperature from each weather record, while the Reducer performed different operations such as finding the maximum, minimum, and average temperature for each year. The analysis identified 2023 as the hottest year in the sample dataset, with a maximum temperature of 43°C. The supplementary tasks also calculated the minimum and average yearly temperatures and identified the highest temperature recorded overall. Overall, this practical demonstrated how MapReduce can efficiently process and analyze large-scale climate datasets using distributed storage, parallel processing, scalability, and fault tolerance.

Github Repo:

<https://github.com/tushar-2606/BDA-SHP-2026>

